

Subjective Listening Tests in URGENT 2026

Selection Phase (Stage II): ACR/CCR Results

mean of 9
per-language ranks

ID	Instr. Rank	ACR									CCR (aggregated one vs rest)									# A C R	# C C R	(#ACR+ #CCR)/2	Initial subj. rank
		EN	HI	FR	IT	JP	ZH	DE	KR	AR	EN	HI	FR	IT	JP	ZH	DE	KR	AR				
1412 WR*	1	3.51	3.91	3.55	3.31	3.45	3.64	3.45	3.49	3.66	+0.11	+0.05	+0.21	+0.08	+0.07	+0.07	0.03	0.06	-0.03	1.78	2.56	2.17	*
1397 GHW*	2	3.47	3.85	3.50	3.14	3.40	3.69	3.49	3.27	3.72	+0.09	+0.00	+0.17	-0.08	+0.07	+0.09	0.06	0.06	-0.07	2.22	3.00	2.61	*
1401 subatomicseer	3	3.35	3.51	3.30	3.32	3.22	3.33	3.33	3.25	3.53	-0.15	-0.30	-0.32	+0.11	-0.19	-0.10	-0.03	-0.07	-0.27	4.11	5.00	4.56	2
1395 baird	4	3.38	3.74	3.17	3.23	3.50	3.74	3.36	3.43	3.72	+0.14	+0.16	+0.20	+0.54	+0.21	+0.19	0.05	0.07	+0.35	2.44	1.22	1.83	1
1348 RICK2000	5	3.35	3.69	2.98	2.77	3.11	3.50	3.09	3.01	3.50	-0.07	+0.06	-0.21	-0.57	-0.20	-0.06	-0.08	-0.10	-0.10	5.00	4.89	4.94	4
1399 Ali-Universal-SE	6	3.28	3.69	2.98	2.93	3.20	3.27	3.05	3.01	3.53	-0.12	+0.02	-0.05	-0.08	+0.04	-0.19	-0.03	-0.02	+0.11	5.33	4.33	4.83	3

In the selection phase, we performed subjective listening tests for the top 6 systems according to the instrumental rank from the qualification phase.

For each language, we determine a ranking, which we average over all languages and the two test methods (ACR/CCR), yielding the “**initial subjective rank**”.

To include a statistical significance analysis into the final ranking, we perform a second consistency check based on pairwise comparison between adjacently ranked systems. We count the amount of statistically significant wins of each pair and switch/retain ranks according to outcome. In case of an equal number of statistically significant wins, the instrumental rank acts as a tiebreaker. Statistical significance is measured according to slide 3 and 4.

The teams marked with an asterisk (*) are not ranked as they were found to be in violation of challenge rules (Multiple registrations from the same team). For completeness, scores of disqualified teams are shown but not included in the official ranking.

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FINAL RANKS (based on most wins in significant comparisons)

x indicates winner system - **green**: significant, **blue**: insignificant

Final rank based on total #wins (sig)

ID	Init. Rank	ACR									CCR									#ACR wins	#CCR wins	Total #WINS	#ACR wins (sig)	#CCR wins (sig)	total #wins (sig)	Final Rank	Instr. Rank
		EN	HI	FR	IT	JP	ZH	DE	KR	AR	EN	HI	FR	IT	JP	ZH	DE	KR	AR								
1395	1	x	x			x	x	x	x	x	x	x	x	x	x	x	x	x	x	7	9	16	5	9	14 →	1	4
1401	2			x	x															2	0	2	1	0	1		3
1401	2	x		x	x	x	x	x	x					x		x				7	2	9	4	1	5	2 ←	3
1399	3		x							x	x	x	x		x		x	x	x	2	7	9	1	4	5		6
1399	3			x	x	x				x			x	x	x			x	x	4	5	9	1	5	6 →	3	6
1348	4	x	x				x	x	x		x	x				x	x			5	4	9	1	2	3 →	4	5
1401	2			x	x	x		x	x	x				x	x		x			6	3	9	4	2	6		3
1348	4	x	x				x				x	x	x			x		x	x	3	6	9	2	4	6	4	5

1. 1395 vs 1401 → 1395 wins with 14:1 significant wins

2. 1401 vs 1399 → significant wins are tied, 1401 wins due to higher instrumental rank

3. 1399 vs 1348 → 1399 wins with 6:3 significant wins

4. Consistency check: 1401 vs 1348 → significant wins are tied, higher instrumental rank places 1401 (2nd rank) higher than 1348 (4th rank)
- 1st rank: 1395

→ 2nd rank: 1401

→ 3rd rank: 1399, 4th rank: 1348

Final ranking: 1395, 1401, 1399, 1348 → 1st: baird, 2nd: subatomicseer, 3rd: Ali-Universal-SE, 4th: RICK2000

Subjective Listening Tests in URGENT 2026

Statistical Analysis in Context of 3GPP GSM Standardization (CCR)

Experiments 4: Influence of Input Level, Voice Activity Detection and Discontinuous Transmission (CCR)

The statistics to be reported from this CCR test are the averaged CMOS (CMOS_k) scores and the standard deviations (S_k) for all the conditions.

Additionally, the requirement in [1, Clause 6.1.4] should be checked using hypothesis tests for the conditions 7-24 if the mean CMOS score is greater than zero (the NS performance is preferred) and greater or equal to zero (the NS performance is equivalent) within a 95 % confidence.

The hypothesis test should be performed using a 1-tailed T-test. The NS algorithm has failed the requirement at level “preferred” for any of test condition if

$$t < t_{N,0.05}$$

where

$$t = \frac{\text{CMOS}_{test}}{S_{test} / \sqrt{N}}$$

and the subscripts $_{test}$ denotes the test condition, N is the number of votes, and $t_{N,0.05}$ is the inverse of the Student’s t-distribution with N degrees of freedom and probability 0.05.

Similarly, the NS algorithm has failed the requirement at level “equal” if

$$t < -t_{N,0.05}$$

[Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); Minimum Performance Requirements for Noise Suppressor Application to the AMR Speech Encoder - ETSI TS 126 077 (2001-03)]

Subjective Listening Tests in URGENT 2026

Statistical Analysis in Context of 3GPP GSM Standardization (ACR)

Experiments 2a, 2b & 2c: No degradation of Speech and no Undesirable Effects in Residual Noise in Conditions with Background Noise

The statistics to be reported from this ACR test are the averaged MOS (MOS_k) scores and the standard deviations (S_k) for all the conditions.

Additionally, the requirement in [1, Clause 6.1.3] should be checked using a hypothesis test for the conditions 25-36 if the mean MOS score is greater or equal to the MOS score for the corresponding equivalent (all being equal except NS activated) reference condition for AMR without NS within a 95 % confidence.

The hypothesis test should be performed using a 2-tailed T-test. The NS algorithm has failed the requirement if, for any of test condition,

$$t < -t_{N,0.05}$$

where

$$t = \frac{MOS_{test} - MOS_{ref}}{\sqrt{\frac{S_{test}^2 + S_{ref}^2}{N}}}$$

and the subscripts $_{test}$ and $_{ref}$ denotes the test condition and corresponding reference condition, respectively, N is the number of votes, and $t_{N,0.05}$ is the inverse of the Student's t-distribution with N degrees of freedom and probability 0.05.

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